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Can vibrotactile stimulation drive brain clearance?

Wearable vibrotactile stimulation, brain rhythms, and glymphatic clearance — rationale, plan, and first electrophysiology (in the mouse)

Pardis Miri · a Snyder Lab (Stanford) × Buzsáki Lab (NYU) collaboration

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Part 1 · The vision — clearing the brain with vibrotactile stimulation

Alzheimer's resists drugs — non-invasive neuromodulation is an open frontier

The strongest non-drug clinical evidence today is multidomain lifestyle (FINGER, US POINTER). Non-invasive stimulation is largely unproven — which is the opportunity.

The proposal: use vibrotactile stimulation to engage brain rhythms and drive clearance

Wearable vibrotactile stimulation → entrain brain rhythms → engage glymphatic clearance of amyloid-β / tau. A mechanism hypothesis — explicitly not yet a treatment claim.

Why vibrotactile, and why wearable: the white space next to light and sound

40 Hz light/sound (GENUS) is the developed approach but hard to wear daily. Vibrotactile stimulation is the open frontier — wearable, sleep-compatible, and pairs naturally with sensing.

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Part 2 · What the literature says — and what to test

40 Hz sensory stimulation lowers amyloid and drives clearance — animals, and early humans

GENUS / Tsai 2016: 40 Hz light/sound lowers amyloid. Murdock 2024 (Nature): 40 Hz gamma drives glymphatic Aβ clearance
Human gamma-sensory is safe with target engagement (Chan 2022).

But a steady-state response is not the same as entraining the brain's own rhythm

A driven 40 Hz response ≠ engaging native gamma (Buzsáki/Soula; some studies find 40 Hz light fails to entrain native gamma in AD-model mice). We separate target engagement from disease modification.

Clearance is actively driven by neural, vascular & sleep dynamics — so 'brain-state' stimulation is plausible

CSF flow tracks NREM slow waves (Fultz 2019) and EEG delta (Hablitz 2019); neuronal dynamics actively direct CSF perfusion (Jiang-Xie–Kipnis 2024, Nature); human sleep clearance moves plasma Aβ/tau (Dagum 2026).

The vibrotactile anchor: 40 Hz vibration helps in mouse AD models, and cortex codes vibration frequency

40 Hz whole-body vibration reduces pathology and improves function in AD-model mice (Suk 2023). Somatosensory cortex faithfully encodes vibration frequency (Romo, Mountcastle) — vibrotactile input can carry a frequency code.

So which vibrotactile parameters should we test?

40 Hz (gamma) · ~50 Hz (our electrophysiology, Part 3) · slow ~1 Hz / sleep-band (clearance biology) · steady vs coordinated-reset patterning · controls: 20 Hz, sham, matched salience.

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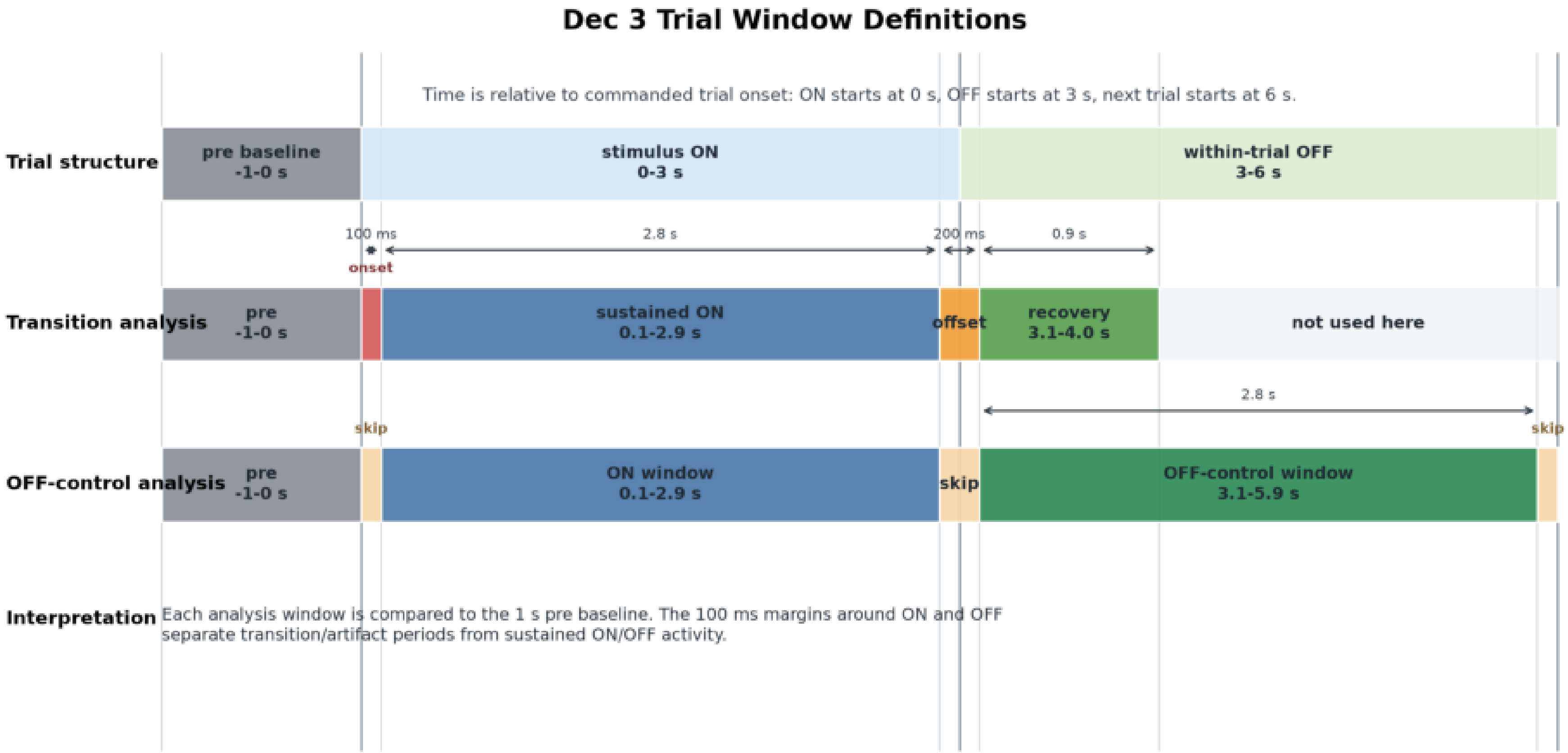
Part 3 · Does vibrotactile stimulation reach the brain? A first study

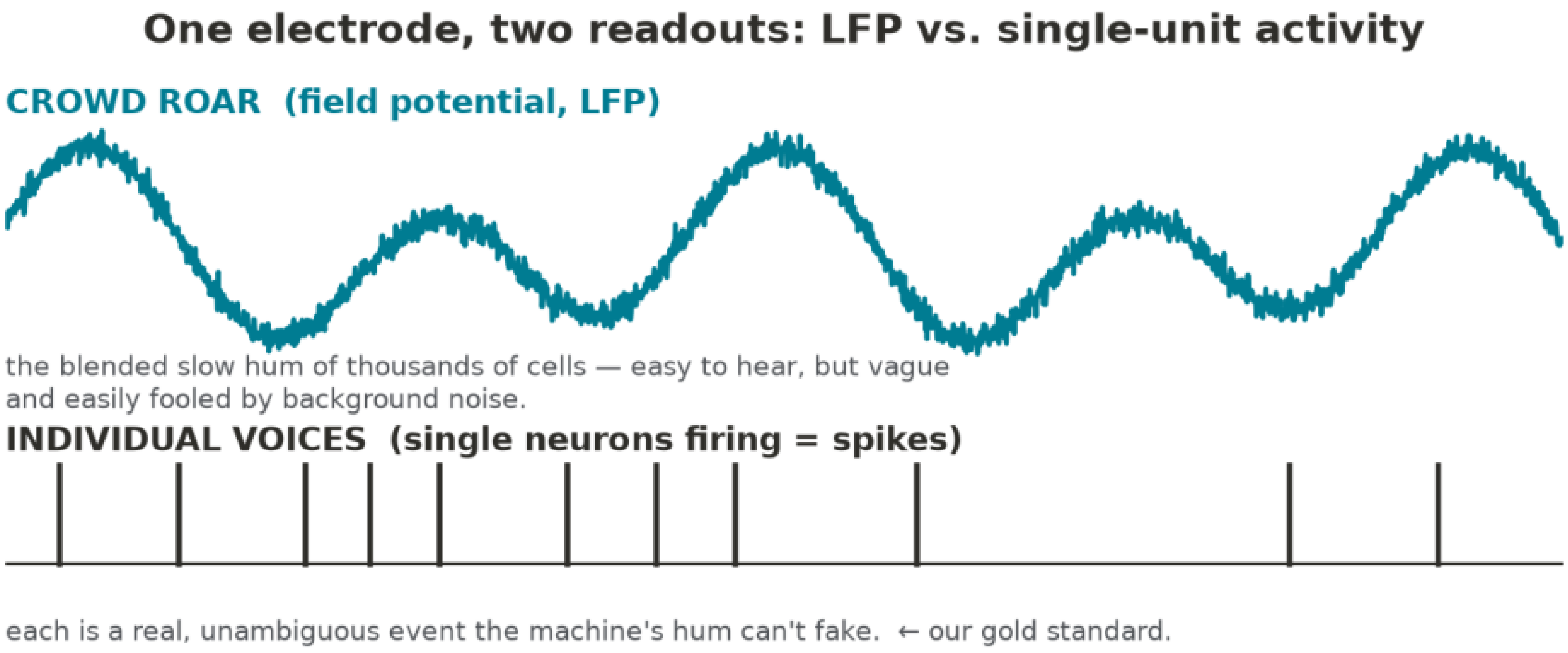
Foundational electrophysiology in the mouse — before any clearance claim

Design: parametric vibrotactile stimulation with simultaneous electrophysiology



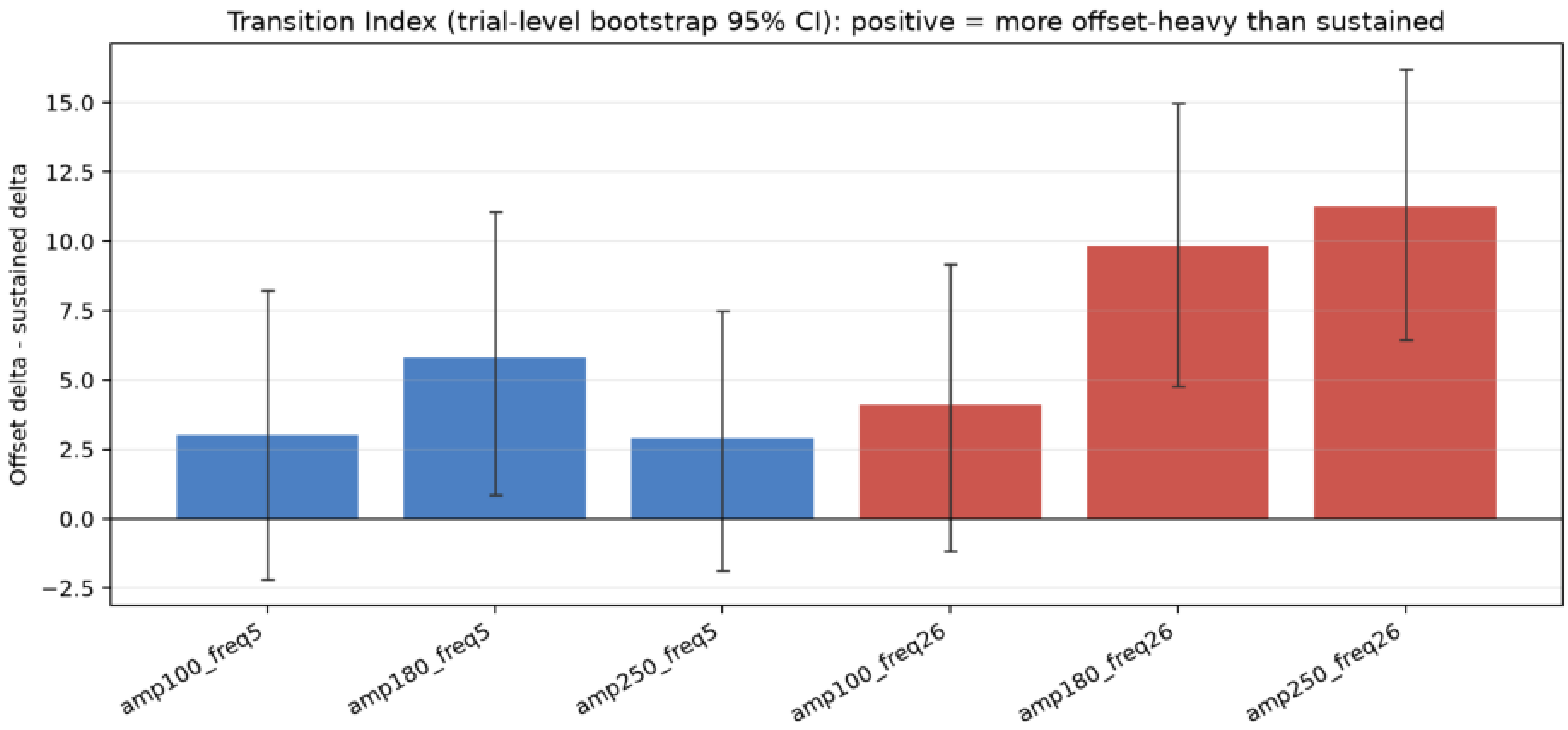
3 s ON / 3 s OFF blocks across four carrier frequencies (5/10/26/50 Hz) and three amplitudes; ~200 repeats/condition; dual-region linear-probe recording (hippocampus + entorhinal cortex).

One electrode yields two readouts: LFP and single-unit activity



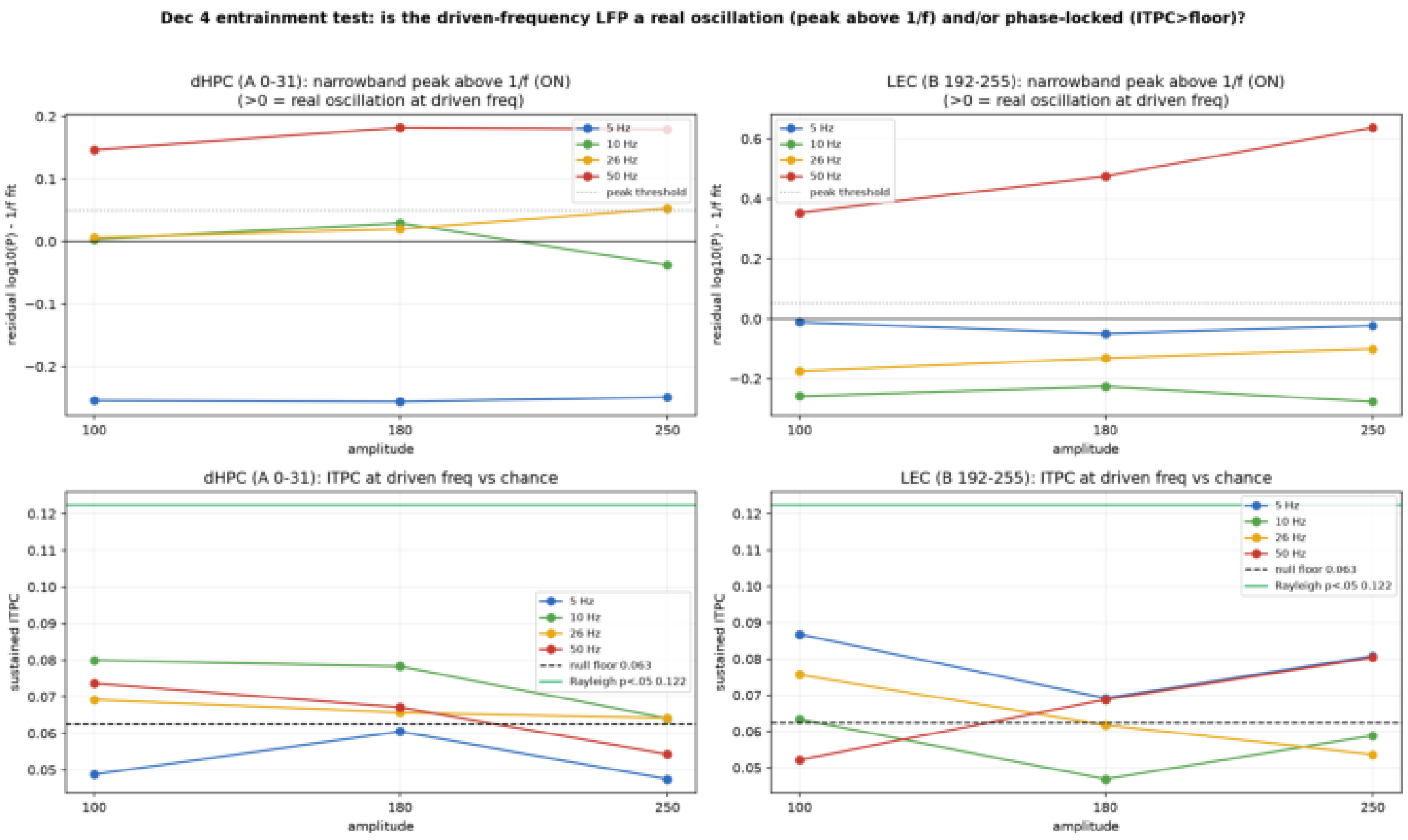
LFP = low-frequency aggregate field (many cells). A single unit = one individual neuron's firing (spikes), isolated by spike sorting. They differ in what can fake them — spikes are the conservative readout.

A robust LFP response — transition-weighted, not a sustained oscillation



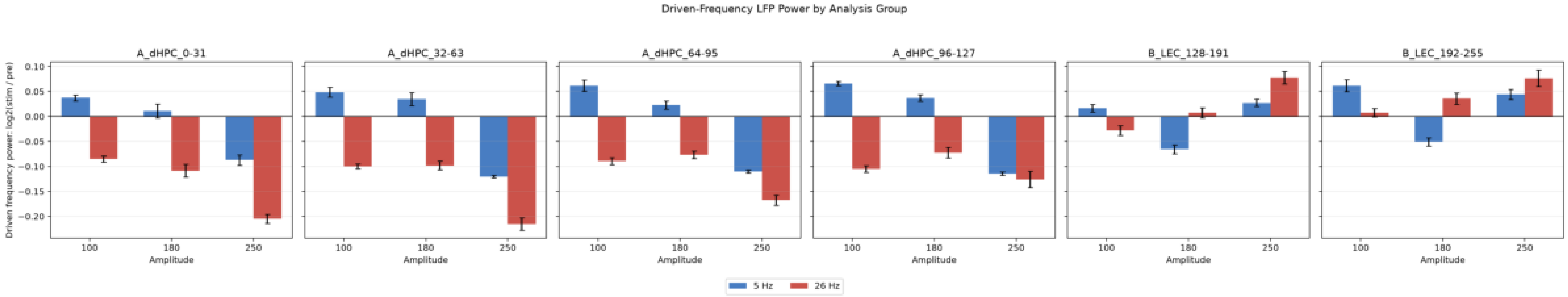
The broadband LFP responds at stimulus onset/offset (an evoked transient), not as a sustained rhythm at the carrier frequency.

No frequency-following in hippocampus at any carrier



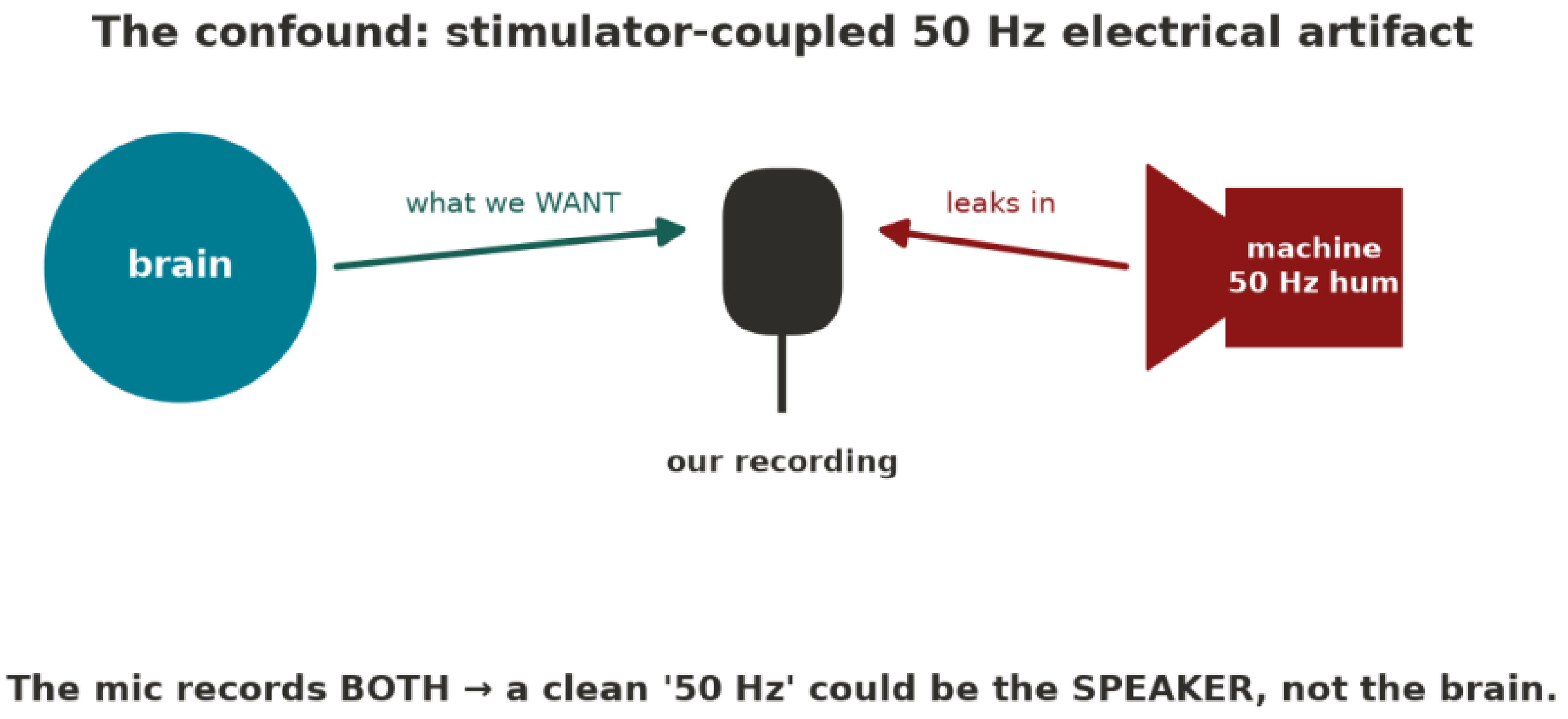
No narrowband spectral peak above the 1/f background, and inter-trial phase consistency at chance — across 5/10/26/50 Hz, replicated across sessions.

Entorhinal cortex shows an amplitude-graded 50 Hz LFP increase



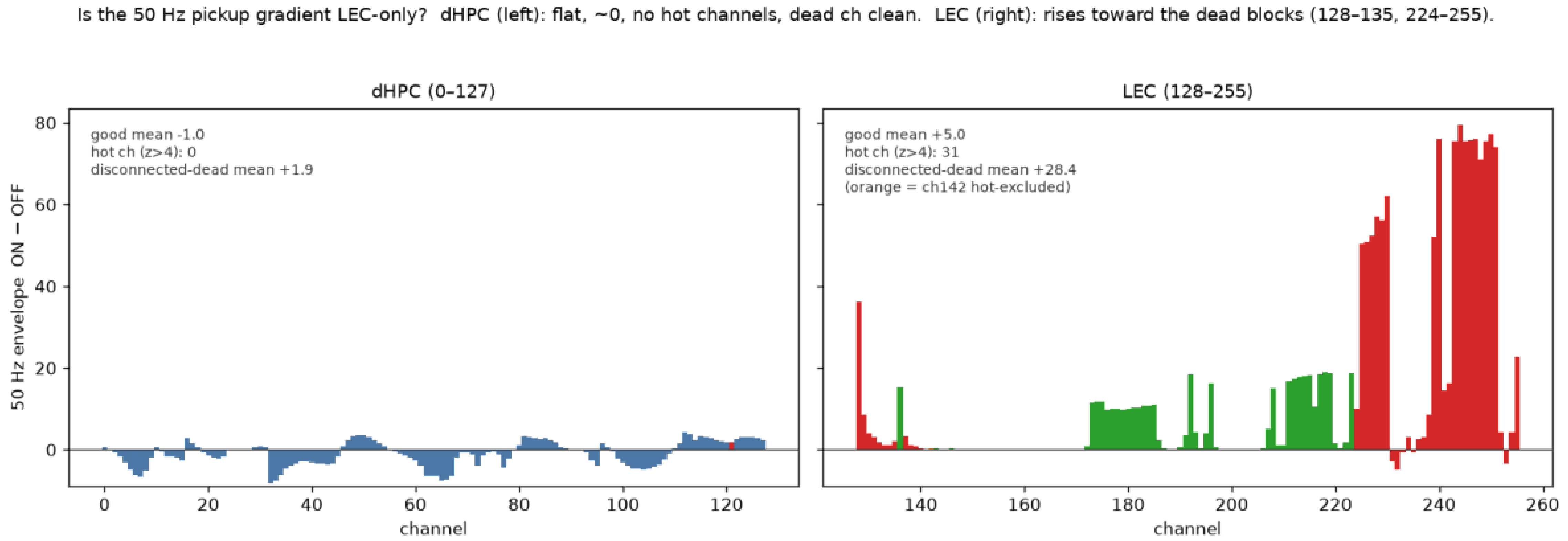
A narrowband 50 Hz LFP power increase, scaling with amplitude — the candidate entrainment signal. But the LFP is exactly the readout most vulnerable to artifact.

The confound: stimulator-coupled 50 Hz electrical artifact



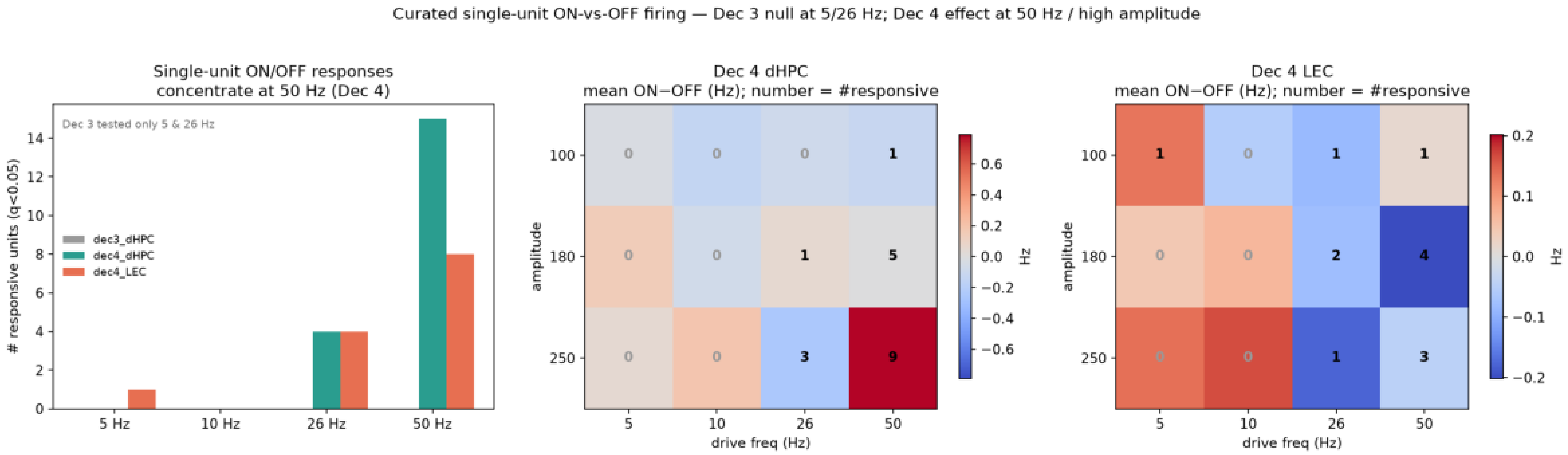
An electromechanical actuator can inject a 50 Hz electrical artifact directly into the recording — indistinguishable from a neural 50 Hz in the LFP alone.

Disconnected channels carry the 50 Hz — so it is largely pickup



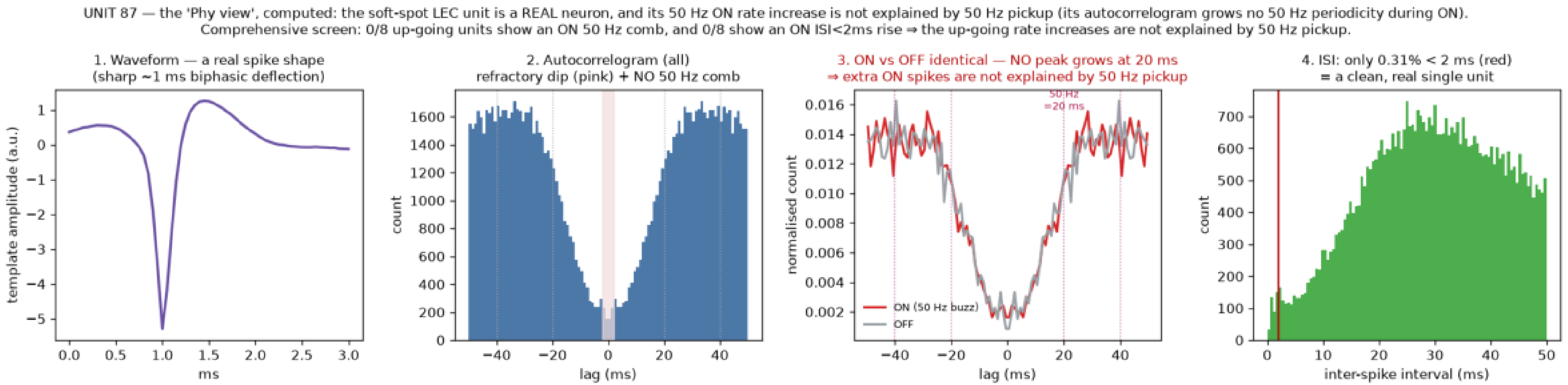
Disconnected electrodes (which cannot record neural signal) show ~6× more 50 Hz than in-tissue channels. The entorhinal 50 Hz LFP is substantially artifact.

Frequency-specific single-unit rate modulation at 50 Hz



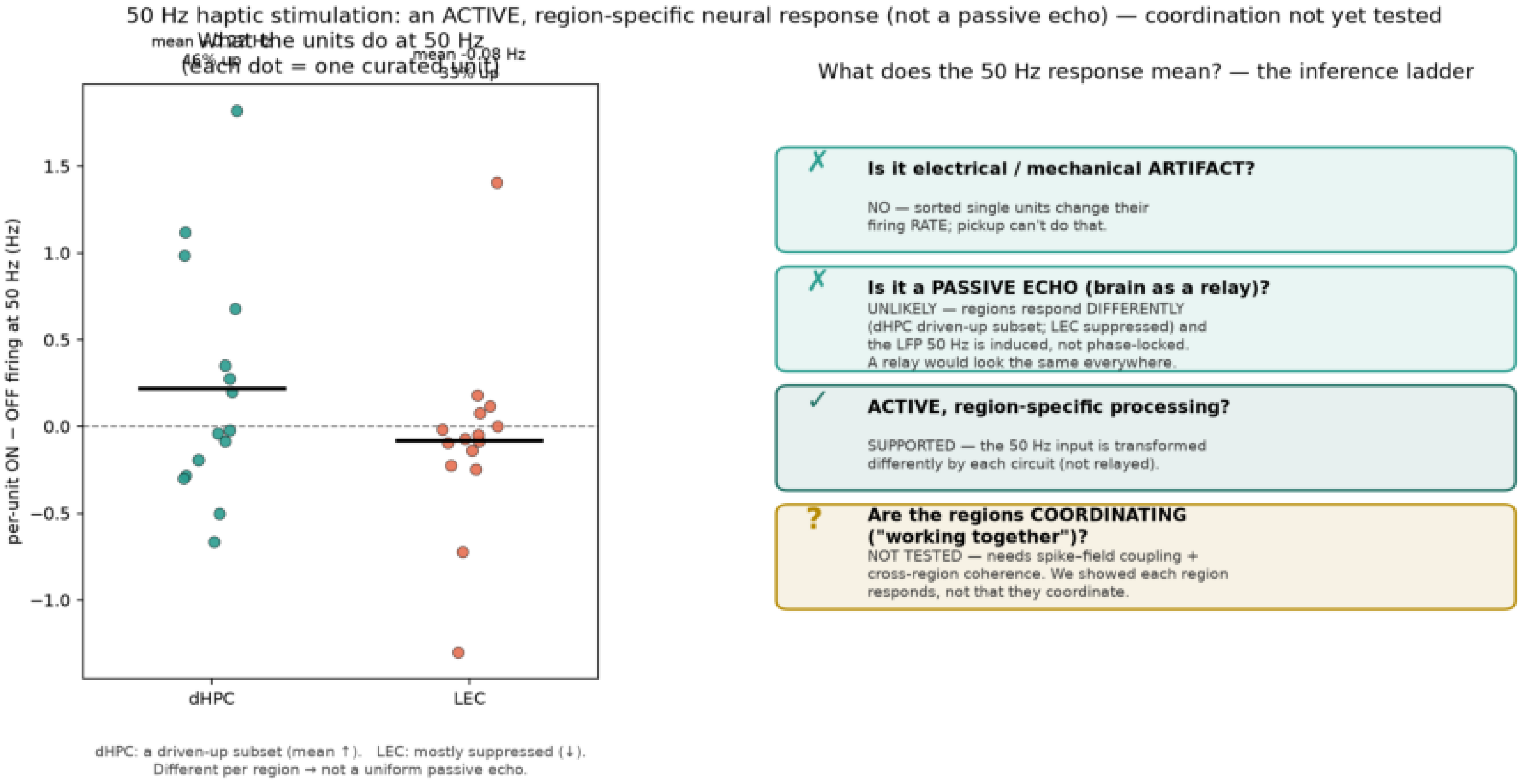
Sorted single units change firing rate during 50 Hz / high-amplitude stimulation, in both regions — an effect electrical pickup cannot produce.

Controls: high-pass separation + autocorrelogram / ISI screens



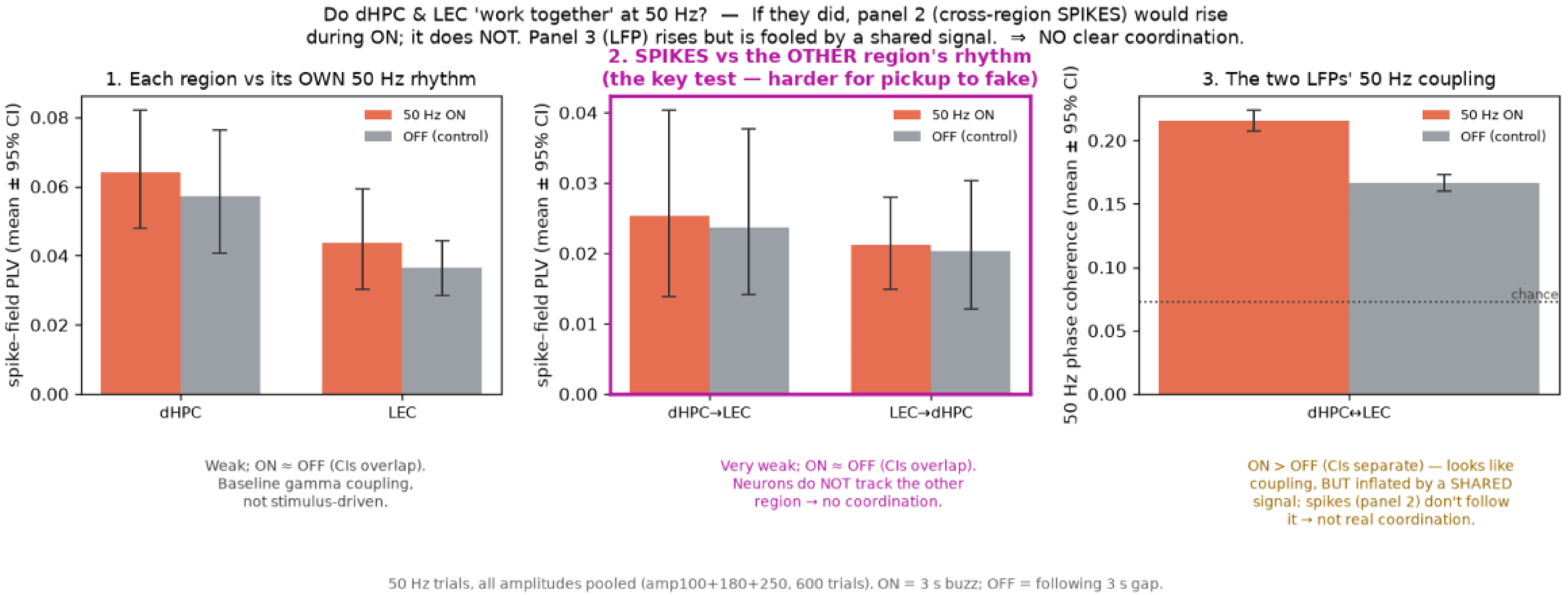
Spikes are detected >~300 Hz (50 Hz pickup is removed pre-detection); units show no stimulus-locked 50 Hz periodicity and no ON-rise in refractory violations.

Region-specific, bidirectional modulation — not a passive relay



Hippocampus: a driven-up subset. Entorhinal cortex: net-suppressed. Opposite transformations of identical input argue for active, circuit-specific processing.

No clear cross-regional coordination at 50 Hz



LFP-LFP coherence rises, but the artifact-resistant cross-region spike-field measure does not — parsimoniously a shared signal, not interaction.

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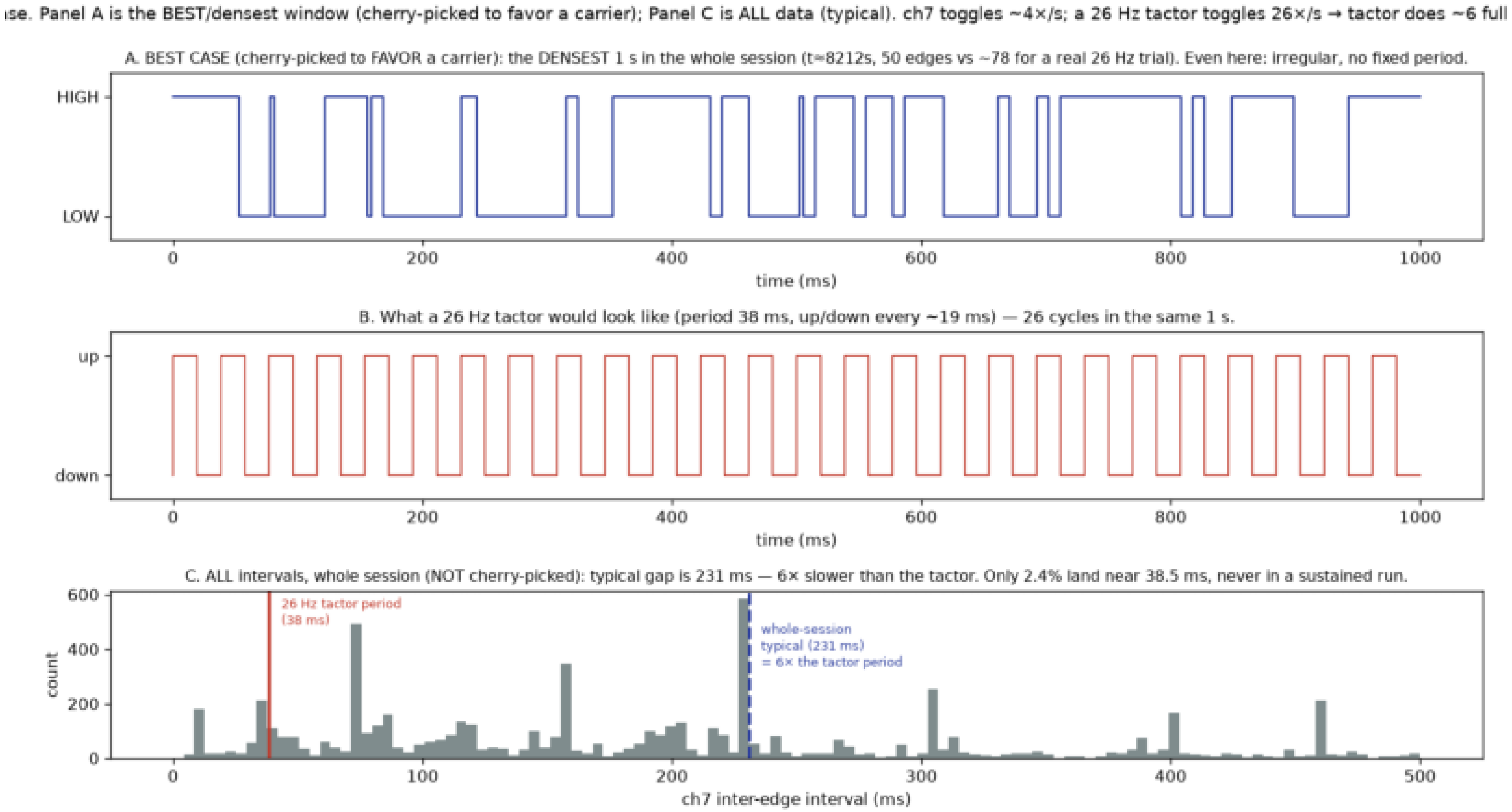
Part 4 · Limits, and the next study

Entrainment was untestable — a measurement gap, not a null

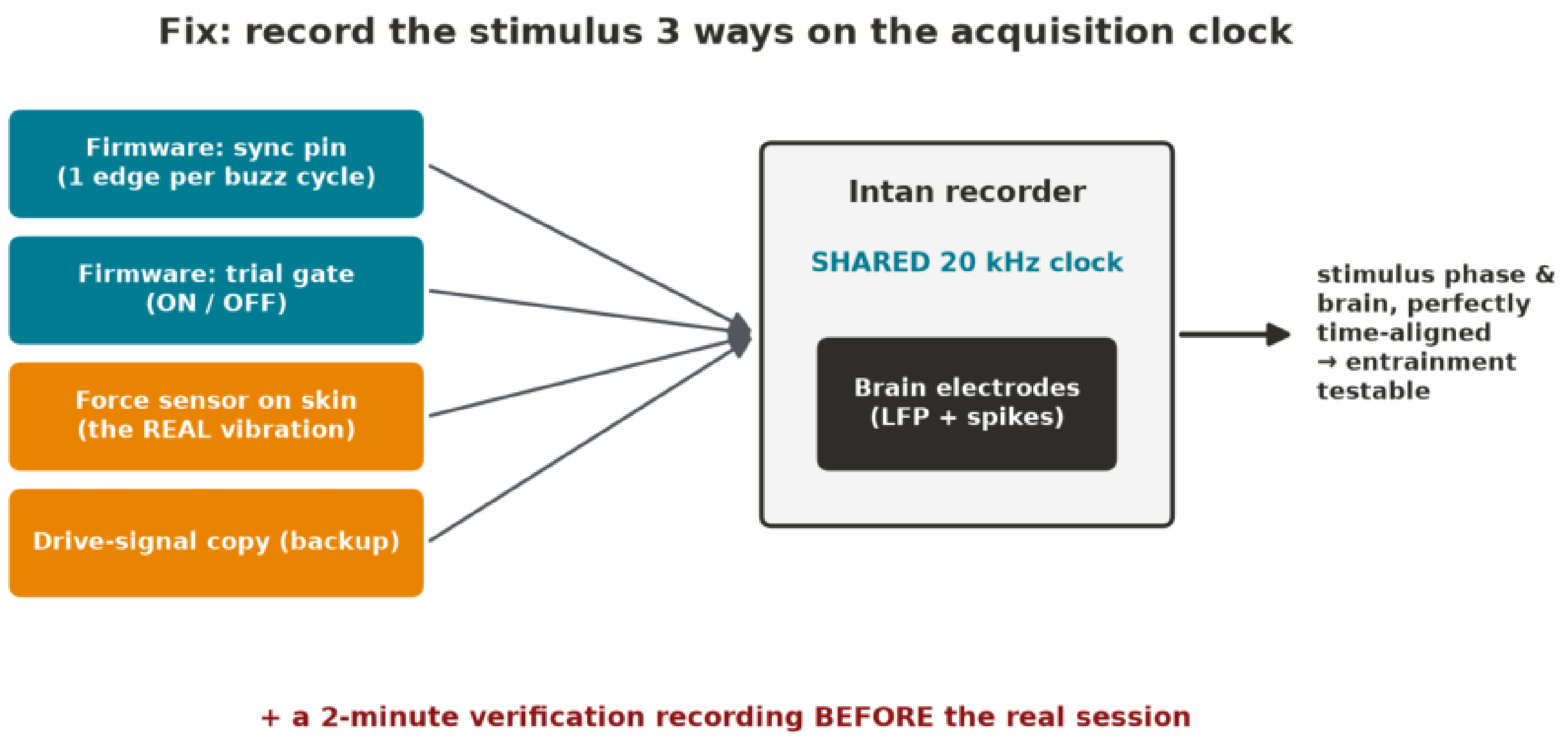
Phase-locking to the stimulus requires a recorded, time-aligned stimulus-phase reference. We lacked one — so entrainment is untestable here, not negative.

The digital sync channel never captured the carrier



The recorded sync line updated at ~4 Hz, independent of carrier (identical pulse counts at 5 vs 26 Hz; ~78 expected at 26 Hz). No phase reference exists in the data.

Fix: redundant, clock-shared stimulus references



Firmware per-cycle sync + a transduced force/accelerometer signal + a drive-signal copy, all on the acquisition clock — cross-checked, with pre-session verification.

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Part 5 · How it fits the program

This study de-risks the program's core premise

It shows peripheral vibration reaches AD-relevant medial-temporal circuits and that those neurons are frequency-tuned (peak ~50 Hz) — directly answering the 'steady-state ≠ engagement' objection.

Key references

GENUS / Tsai 2016 · Murdock 2024 (gamma → glymphatic) · Suk 2023 (40 Hz tactile) · Jiang-Xie–Kipnis 2024 (neural dynamics → CSF) · + Buzsáki / Soula (steady-state ≠ entrainment).

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Thank you — questions

Happy to go into any figure, the artifact controls, the parameter plan, or the next-round instrumentation.